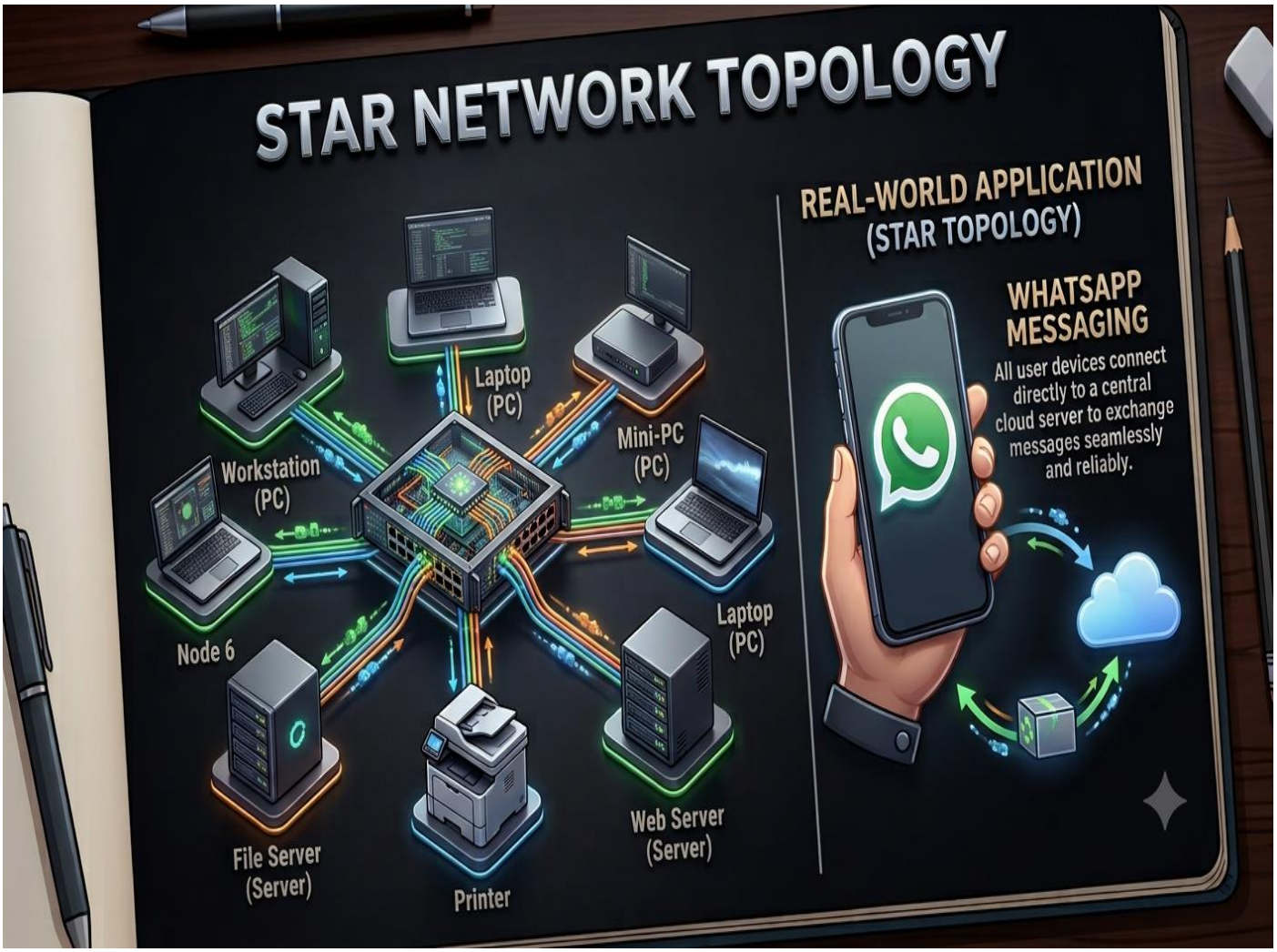
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Assignment-2

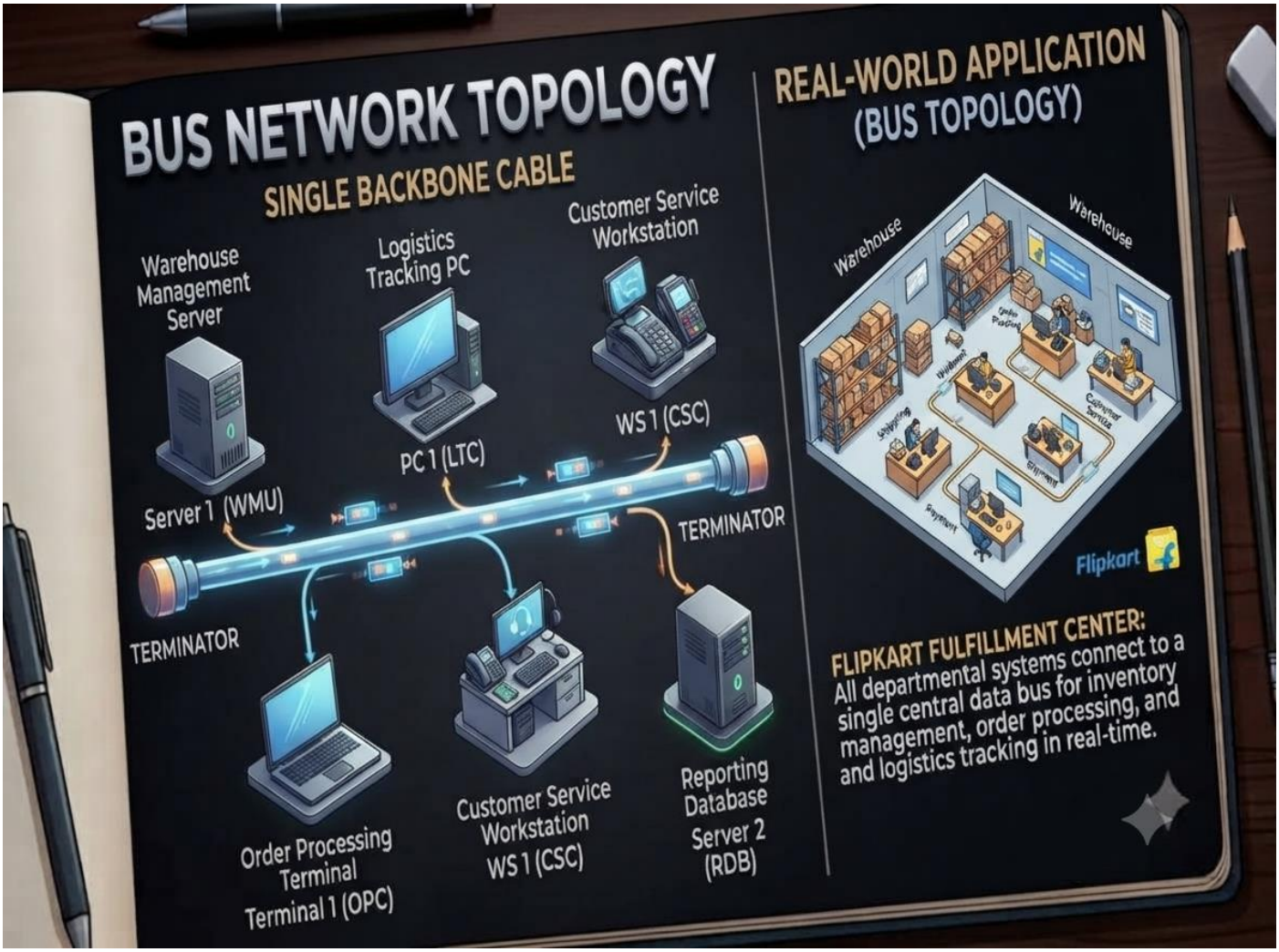
Created By
~pintu mer

1. **Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.**



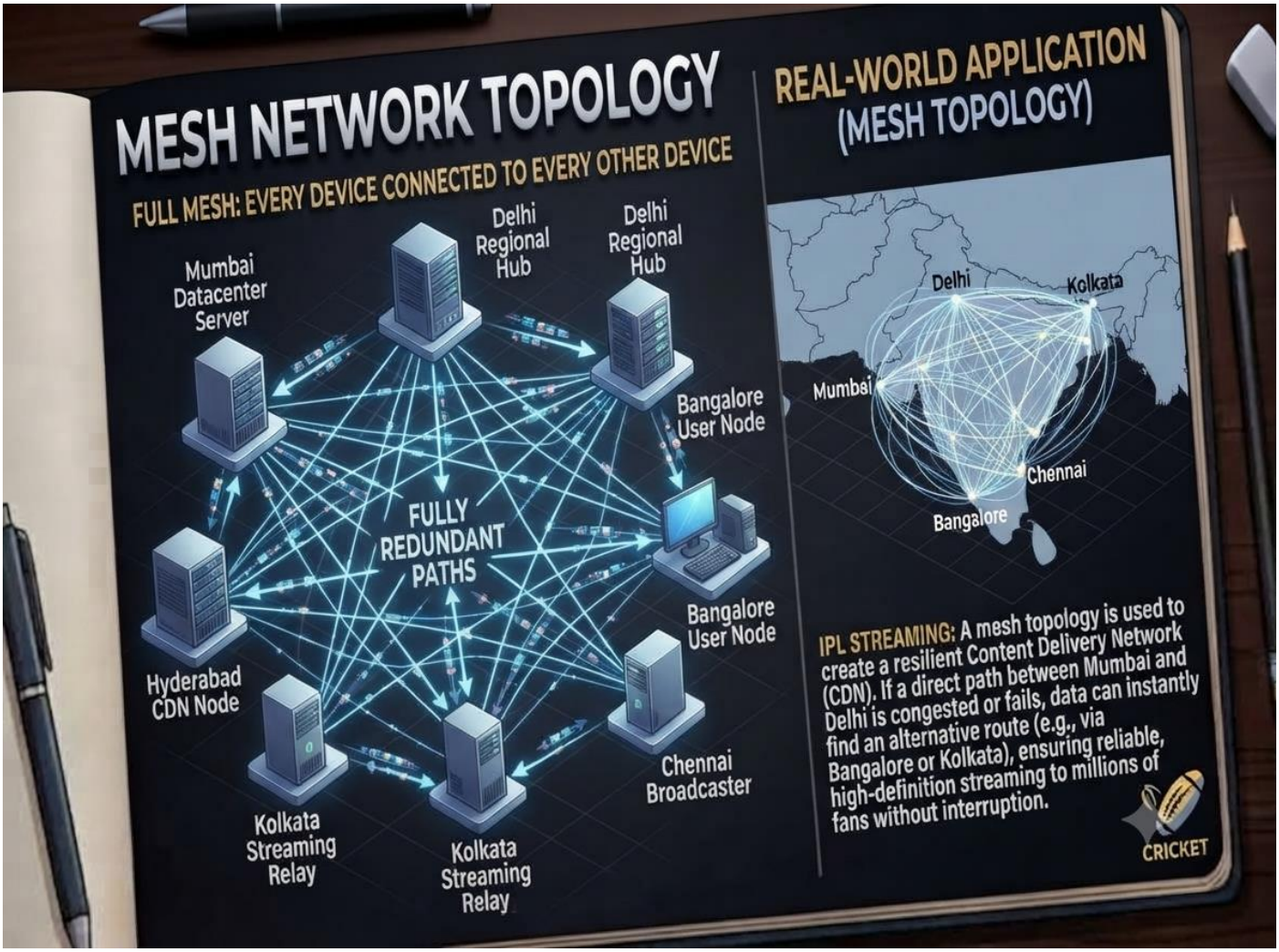
How used this TopoLogy :

- **Centralized Server: WhatsApp uses a central cloud server that acts like the main hub.**
- **Direct Connections: Every user's phone connects directly to this single central server, forming a star pattern.**
- **Fast & Reliable Messaging: When you send a message, it goes straight to the central server first, which then instantly delivers it to the receiver.**
- **Easy Management & Security: Having one central point makes it much easier for WhatsApp to manage heavy traffic, control security, and handle end-to-end encryption.**



How use this Topology :

- Single Backbone Cable (Main Bus): All systems in a Flipkart fulfillment center are connected to a single central cable.
- Connected Departments: Departments like Warehouse Management, Order Processing, Logistics Tracking, and Customer Service are linked to this main bus.
- Data Flow: When data (like an order or inventory update) is sent, it travels across this main backbone cable.
- Target Acceptance: All systems can see the data, but only the specific department targeted for the data accepts and processes it



How use this Topology :

- **Multiple Pathways (Redundancy): Every server and node is connected to every other device, providing multiple paths for data to travel.**
- **No Single Point of Failure: If one direct path or server fails or gets congested, data instantly finds an alternative route (e.g., through other servers).**
- **High Reliability: For events like IPL streaming with millions of fans, it ensures continuous, high-definition video streaming without any interruptions.**

2. Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

Topology	Pros (4)	Cons (4)	Best Use Case	App/Service Example
Star	1. Easy to install 2. Easy to manage 3. Easy to find faults 4. Failure of one cable affects only one device	1. Needs more cable 2. Needs a central switch/hub 3. Switch failure affects the network 4. Can be costly for large networks	Office / College computer lab	Instagram – users connect to central servers to access services.
Bus	1. Simple design 2. Uses less cable 3. Low cost4. Easy for small networks	1. Main cable failure affects the network 2. Difficult to troubleshoot 3. Performance decreases with more devices 4. Data collisions can occur	Small/temporary networks	Zomato – a simple shared network can be used for small internal systems.
Ring	1. Data moves in an organized way 2. Fewer data collisions 3. Equal access for devices 4. Good for predictable traffic	1. One break can affect communication 2. Adding/removing devices can be difficult 3. Troubleshooting can be harder 4. Less flexible than star	Networks requiring orderly data flow	Paytm – reliable communication paths can support transaction systems.
Mesh	1. Very reliable 2. Multiple paths for data 3. One link failure may not stop the network 4. Good fault tolerance	1. Expensive 2. Requires many cables/connections 3. Complex to install 4. Difficult to manage at large scale	Critical/high-availability networks	Instagram – multiple network paths can help keep services available during heavy traffic.

3. **Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.**

❖ Network Topology for Multiplayer Game: Star Topology

- **Topology: Star Topology**
- **Speed: Provides high speed and low latency, which is important for real-time multiplayer gaming.**
- **Reliability: If one game server fails, other servers continue working because they are connected separately.**
- **Cost: Higher cost because a powerful central switch and additional network equipment are required.**
- **Why suitable: It is easy to manage, expand, and troubleshoot, making it suitable for games like PUBG or Free Fi**

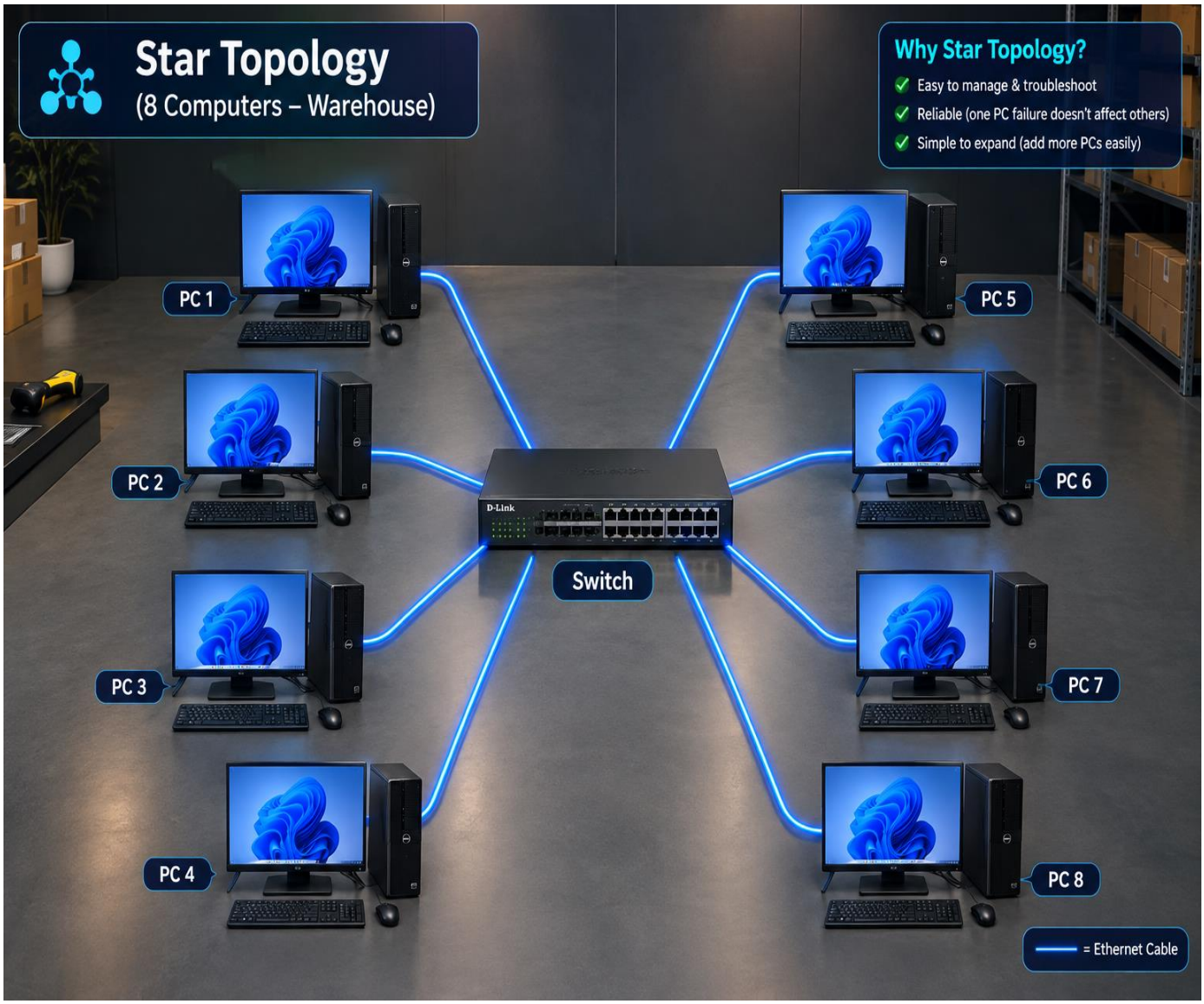
4. **Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.Hint:Consider factors like cost, ease of troubleshooting, and future expansion.**

❖ Recommended Topology: Star Topology

Reason:

- **Easy to manage and troubleshoot: If one computer or cable fails, other computers continue working.**
- **Future expansion: More computers can easily be added to the switch.**
- **Cost: Requires a central switch and cables, but it is cost-effective and reliable for a small warehouse.**

➢ Dia-gram :


Star Topology
(8 Computers – Warehouse)
Why Star Topology?
Easy to manage & troubleshoot
Reliable (one PC failure doesn't affect others)
Simple to expand (add more PCs easily)
PC 1
PC 2
PC 3
PC 4
PC 5
PC 6
PC 7
PC 8
D-Link
Switch
= Ethernet Cable